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of PUBLIC HEALTH

Basic Mapping of Epidemiologic Data

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Learning Objectives

- ▶ Describe the role of geography in health
- ▶ Explain what geographic information systems (GIS) are
- ▶ Describe options of GIS software tools
- ▶ Create a basic choropleth map using QGIS



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Geography and Health

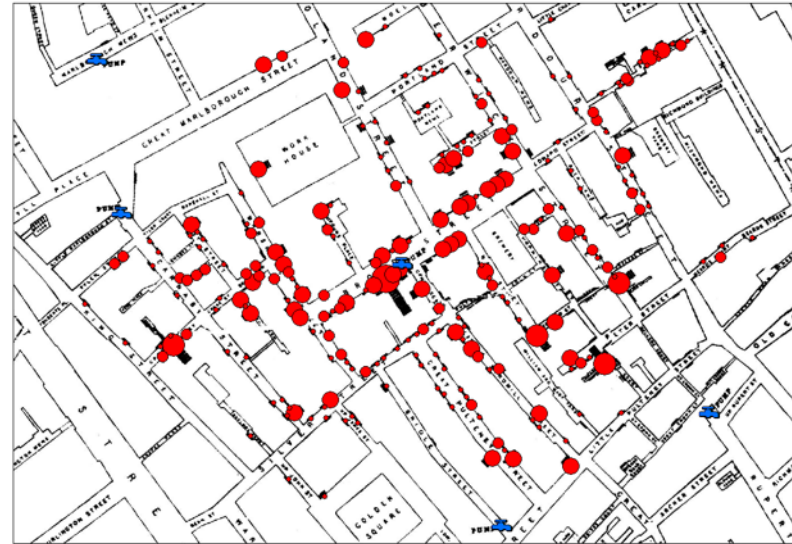
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Does “Place” Matter?

- ▶ Why collect location information for health, disease, and wellness?
- ▶ Our environment has a significant influence on our health and well-being
- ▶ Utilize a wide interpretation of what our “environment” is:
 - ▶ Physical (air, ground, water)
 - ▶ Built (person-made structures that surround us)
 - ▶ Behavioral/social (crime, poverty, etc.)
 - ▶ Food/beverages (access)
 - ▶ Media (advertisements)

Place Matters—Disease Distribution

- ▶ It has long been recognized that geography influences illness and disease
- ▶ Certain diseases occur at a higher rate in some places and not in others
- ▶ As far back as the founding of epidemiology as a discipline by Dr. John Snow, a hand-drawn map pinpointed the source of a disease epidemic



Place Matters—Program Planning

- ▶ Local health officials, health service delivery organizations, and health program planners all think about the physical layout of their target communities as one important aspect of program delivery
- ▶ Helps ensure targeted allocation of resources, given that these resources are in limited supply

Obtaining Geographic Data

- ▶ Collected in a study setting
- ▶ Created digitally
 - ▶ Satellite maps
- ▶ Publicly available sources
 - ▶ US Census
 - ▶ Environmental protection agency
 - ▶ Local governments (Departments of Urban Planning, Recreation and Parks, etc.)
 - ▶ GIS data portals

What Are We Looking for in Geographic Data?

- ▶ Are “events” occurring closer together in one area versus another?
 - ▶ Clustering
- ▶ Does risk of a disease vary across geographies?
- ▶ Are events of one type co-located with another?
 - ▶ Example: increase rates of property crime around neighborhoods with higher median income
- ▶ Does exposure in one area allow prediction of exposure in another (where we might not have specific data)?
 - ▶ Example: air pollution



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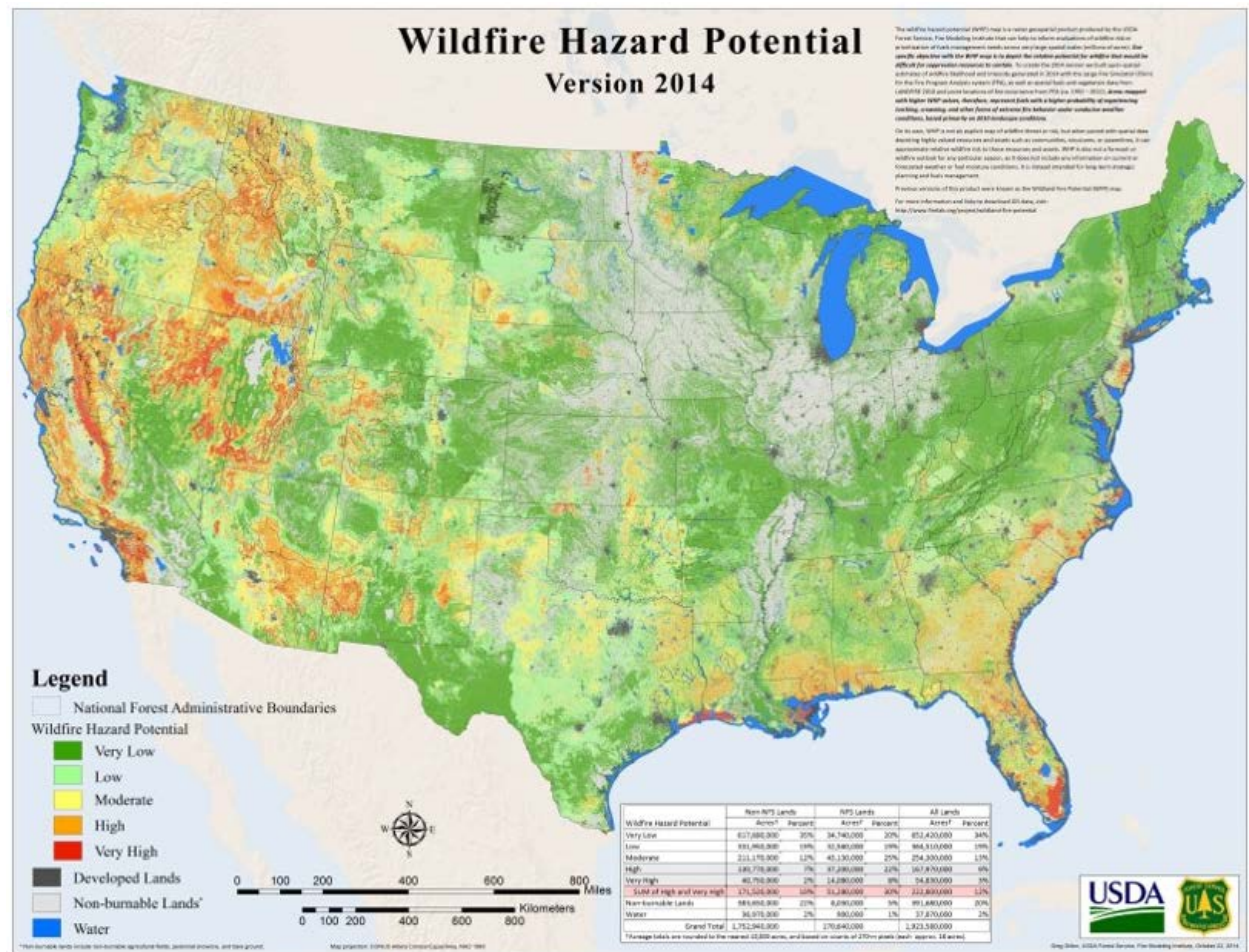
Introduction to Geographic Information Systems (GIS)

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What Is GIS?

- ▶ A tool for integrating and visualizing spatial data from multiple sources
- ▶ A visual database for spatial data
- ▶ It allows...
 - ▶ Storage
 - ▶ Manipulation
 - ▶ Display
- ▶ What can we do with GIS
 - ▶ Show where things occur
 - ▶ Show how much of something occurs in one area versus another
 - ▶ Show change over time and place

Example: Completed Map



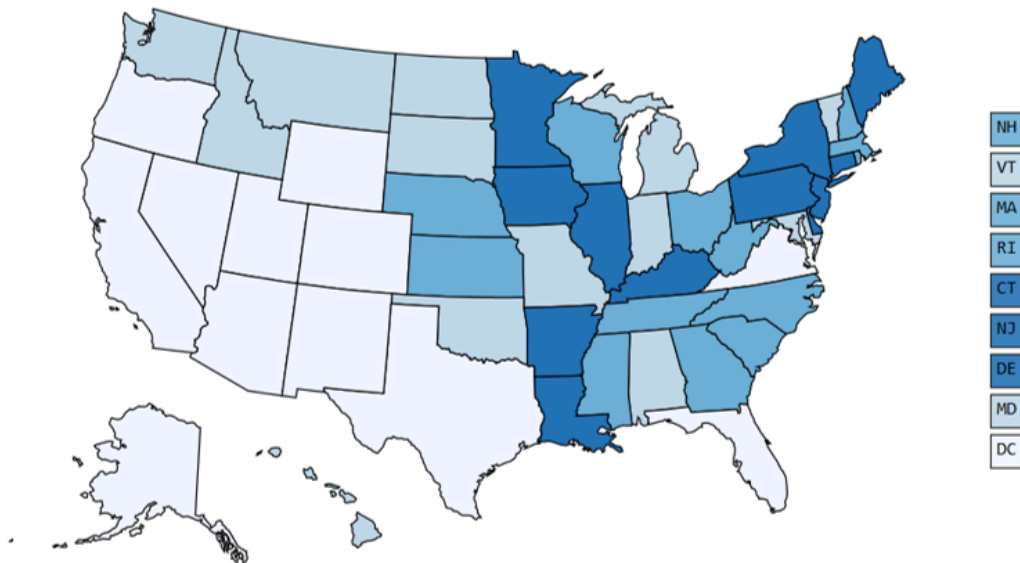
Source: Rocky Mountain Research Station Fire, Fuel, Smoke Science Program. (2015).
Classified 2014 WHP: GIS Data and Maps. Retrieved October 9, 2018.

Example: Interactive Map

Rate of New Cancers in the United States

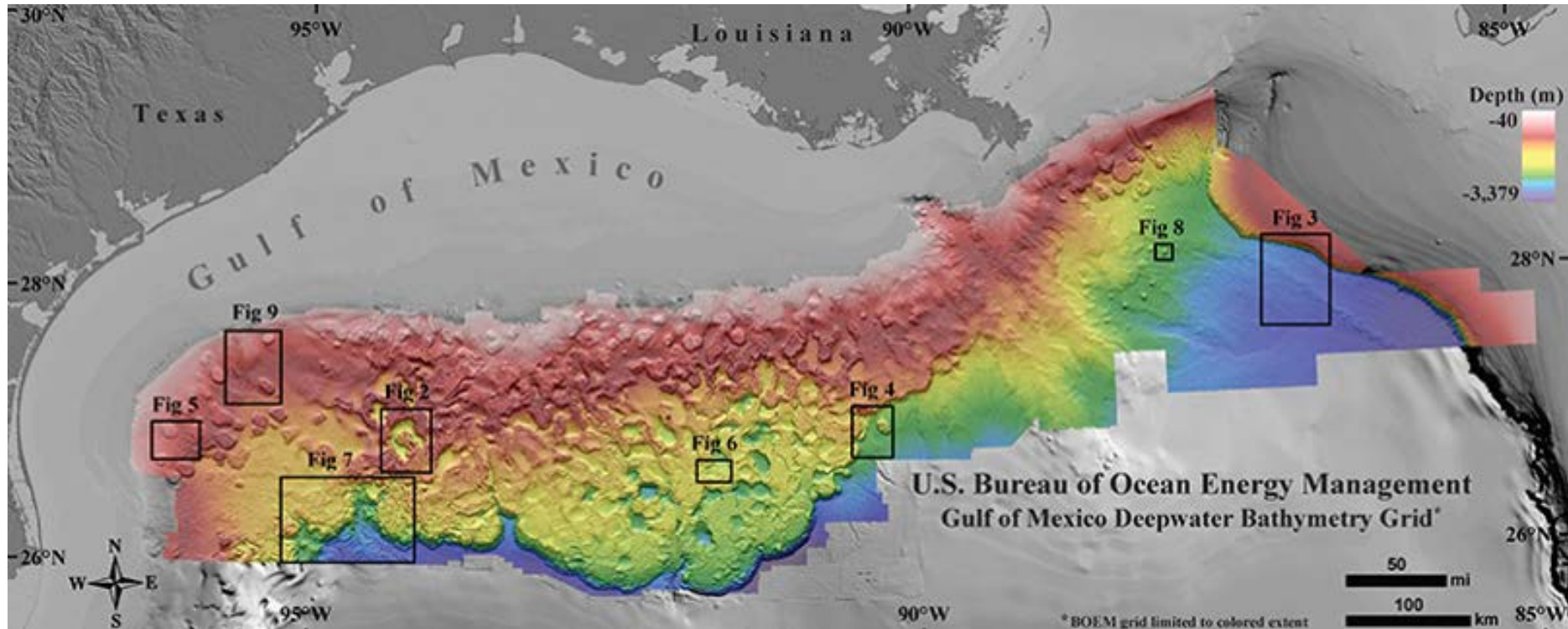
All Types of Cancer, All Ages, All Races/Ethnicities, Male and Female
Rate per 100,000 people

Map Table Chart Export



Source: Centers for Disease Control and Prevention (CDC). [United States Cancer Statistics: Data Visualizations: Rate of New Cancers in the United States](#). Retrieved October 9, 2018.

Example: Three-Dimensional Map



Source: US Department of Interior, Bureau of Ocean Energy Management. (n.d.). Oil and Gas Energy: BOEM Northern Gulf of Mexico Deepwater Bathymetry Grid from 3D Seismic. Retrieved October 10, 2018.

GIS Tools

- ▶ GIS is a generic name for a tool (or software)
 - ▶ There are numerous examples

- ▶ Proprietary options (license required)
 - ▶ ArcGIS (Esri)
 - ▶ Geomedia (Hexagon Geospatial)
 - ▶ MapInfo Professional (Pitney Bowes)
 - ▶ Etc.

- ▶ Free software
 - ▶ QGIS
 - ▶ gVSIg
 - ▶ GeoDa
 - ▶ Etc.

QGIS—1

- ▶ For this course, the exercise will be done using QGIS
- ▶ QGIS: A free and open source geographic information system home page is linked on the lecture page
- ▶ You are welcome to do the exercise using any GIS software program
 - ▶ There are many similarities in how they work
 - ▶ Most have several online help tutorials



QGIS—2

- ▶ Download QGIS (or another GIS software) onto your computer now



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GIS Software

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Types of Spatial Data

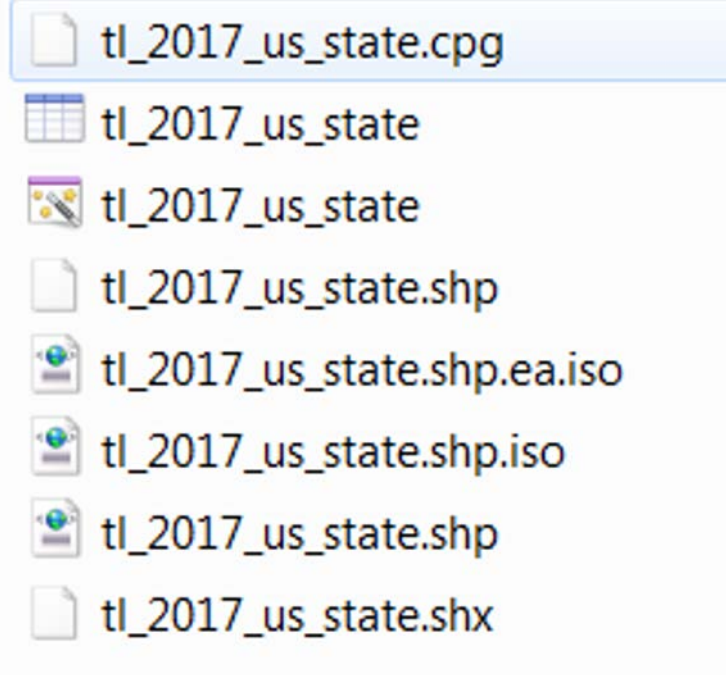
- ▶ Raster images:
 - ▶ Made of pixels (a single point in an image display)
 - ▶ Can be discrete (each pixel has a specific value) or continuous (changes are gradual)
- ▶ Vector images:
 - ▶ Mathematically calculated from one point to another to make a line or shape
 - ▶ Can be given in two-dimensional or three-dimensional space
 - ▶ Vectors can be points, lines, or polygons (shapes)

Shapefiles

- ▶ Format for storing digital vector data
- ▶ Made up of a collection of files (all necessary), including...
 - ▶ .shp: the file that stores the feature geometry
 - ▶ .shx: index of the geometry
 - ▶ .dbf: database of attributes
 - ▶ .prj: the coordinate system and projection information

Shapefile: Example

- ▶ For the exercise for this course, we will be downloading a shapefile of the US states
- ▶ When downloading the file “tl_2017_us_state,” a folder is downloaded containing all of the shown files
- ▶ When organizing files, ensure shapefiles are kept in folders with filenames that do not have spaces or punctuation and are not too lengthy



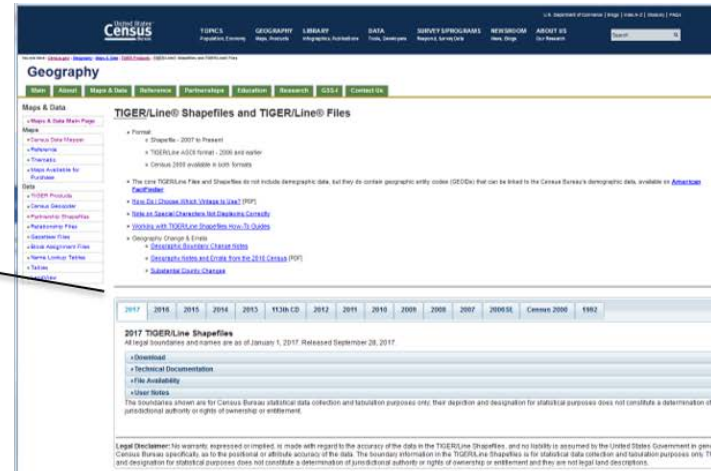
Software: Appearance

- ▶ Software programs have a fairly “typical” appearance, with toolbars and buttons across the top and pull-down menus
- ▶ Maps are created by laying down shapefiles in “layers”:
 - ▶ Layer 1: point locations of events (i.e., crime, residences reporting a disease, etc.)
 - ▶ Layer 2: lines of major roads
 - ▶ Layer 3: polygons of county boundaries
- ▶ Layers can be related to each other using topology:
 - ▶ How many events occurred in each county?

Shapefile for US States—1

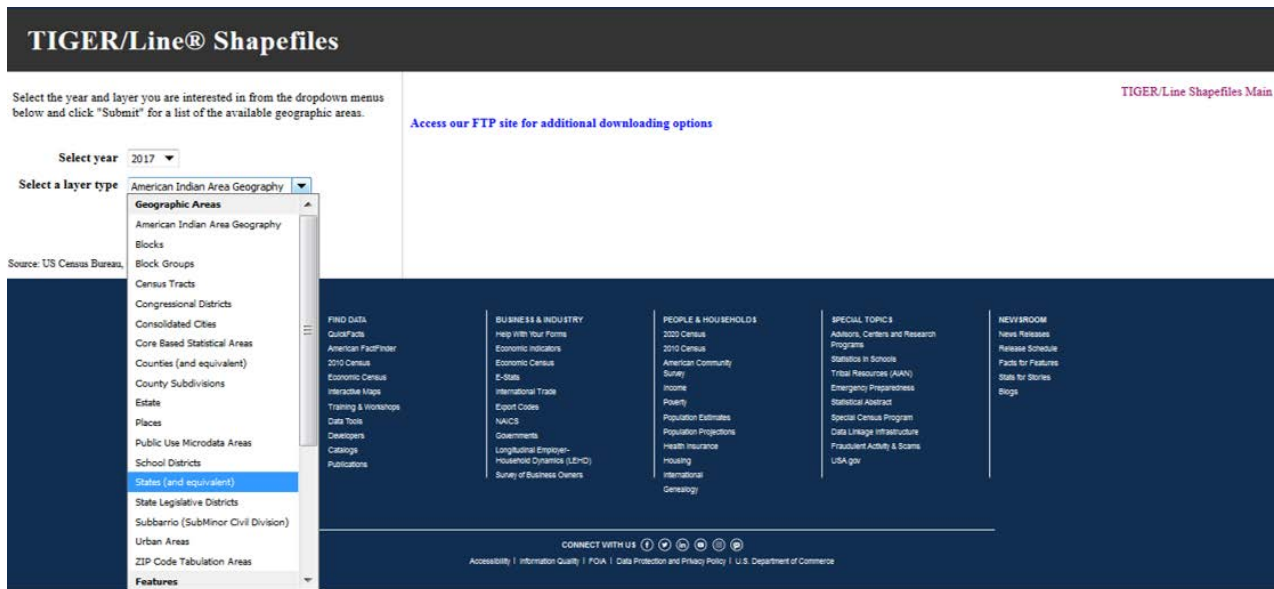
- ▶ The US Census Bureau has a series of TIGER/Line shapefiles
 - ▶ The web address is provided in a link on the lecture page
- ▶ This example uses 2017 data—you can select any year of interest to you

Select 2017 Download,
Web Interface



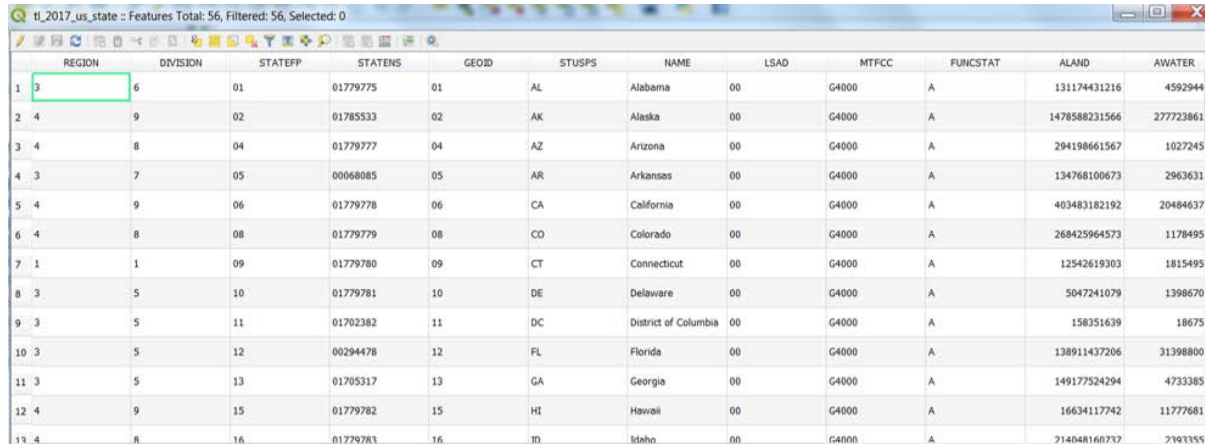
Shapefile for US States—2

- ▶ Select “States (and equivalent)” from the pull-down menu
 - ▶ Then select “download national file”
- ▶ Save the zip file to your folder of choice
 - ▶ Unzip it into a folder with the same name (tl_2017_us_state)



Attribute Table

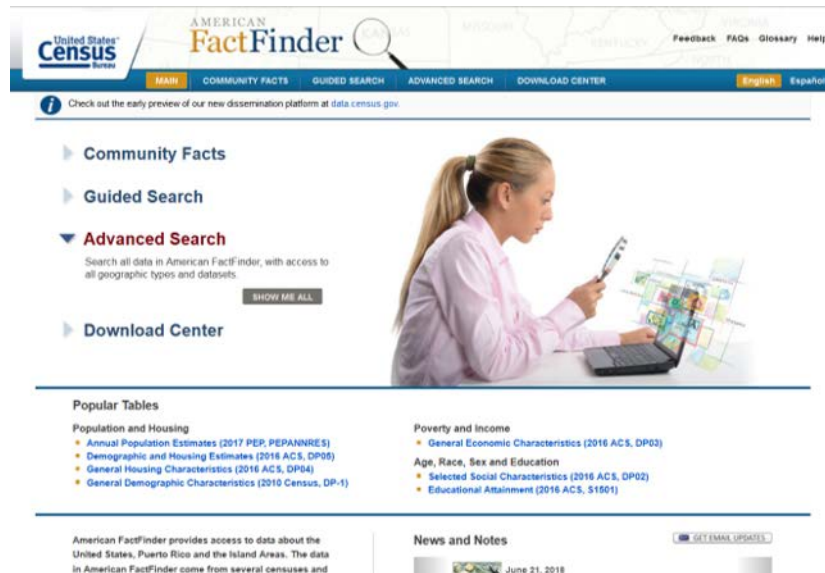
- ▶ This is the “data table” that contains the information included in that layer
- ▶ Attribute table for the shapefile of US states
 - ▶ All TIGER/Line shapefiles contain a geographic entity code that links to the US Census demographic data



	REGION	DIVISION	STATEFP	STATENS	GEOID	STUSPS	NAME	LSAD	MTFCC	FUMSTAT	ALAND	AWATER
1	3	6	01	01779775	01	AL	Alabama	00	G4000	A	131174431216	4592944
2	4	9	02	01785533	02	AK	Alaska	00	G4000	A	1478580231566	277723861
3	4	8	04	01779777	04	AZ	Arizona	00	G4000	A	294198661567	1027245
4	3	7	05	00068085	05	AR	Arkansas	00	G4000	A	134768100673	2963631
5	4	9	06	01779778	06	CA	California	00	G4000	A	403483182192	20484637
6	4	8	08	01779779	08	CO	Colorado	00	G4000	A	268425964573	1178495
7	1	1	09	01779780	09	CT	Connecticut	00	G4000	A	12542619303	1815495
8	3	5	10	01779781	10	DE	Delaware	00	G4000	A	5047241079	1398670
9	3	5	11	01702382	11	DC	District of Columbia	00	G4000	A	158351639	18675
10	3	5	12	00294478	12	FL	Florida	00	G4000	A	138911437206	31398800
11	3	5	13	01705317	13	GA	Georgia	00	G4000	A	149177524294	4733385
12	4	9	15	01779782	15	HI	Hawaii	00	G4000	A	16634117742	11777681
13	4	R	16	01779783	16	ID	Idaho	00	G4000	A	214048160737	2393355

Download US Census Demographic Data—1

- ▶ The American FactFinder is the source from which Census data can be extracted
 - ▶ The web address is provided in a link on the lecture page
- ▶ Select “Advanced Search”
 - ▶ Then select “Show me all”



Download US Census Demographic Data—2

- ▶ For this example, we will use 2017 population data
 - ▶ You can choose to download any data that is available through the US Census at the state level
- ▶ Under “Topic,” select “People,” “Basic Counts and Estimates,” “Population Total”
- ▶ Under “Topic,” select “Year,” “2017”
- ▶ Under “Geographies,” select “State—040,” “All States within the United States and Puerto Rico,” “Add to your Selections”
- ▶ Download and unzip the indicated file

The screenshot shows the American FactFinder interface. On the left, the 'Your Selections' panel is circled in red, showing 'People: Basic Counts/Estimate: Population Total', 'Year: 2017', and 'State: All States within United States and Puerto Rico'. Below this, the 'Search using the options below' section has 'Topics' selected, with 'PS - PINRES' checked in the 'Topics' list. The main search results area shows a table of results. The first row is selected, and its checkbox is also circled in red. The table columns are ID, Table, File or Document Title, Dataset, and About. The selected row is for 'PEPAGESEX' with the title 'Annual Estimates of the Resident Population for Selected Age Groups by Sex for the United States, States, Counties, and Puerto Rico Commonwealth and Municipalities: April 1, 2010 to July 1, 2017'.

ID	Table, File or Document Title	Dataset	About
PEPAGESEX	Annual Estimates of the Resident Population for Selected Age Groups by Sex for the United States, States, Counties, and Puerto Rico Commonwealth and Municipalities: April 1, 2010 to July 1, 2017	2017 Population Estimates	?
PS - PINRES	Annual Estimates of the Resident Population: April 1, 2010 to July 1, 2017	2017 Population Estimates	?
PEPASRSH	Annual Estimates of the Resident Population by Sex, Age, Race Alone or in Combination, and Hispanic Origin for the United States and States: April 1, 2010 to July 1, 2017	2017 Population Estimates	?
PEPASRSH	Annual Estimates of the Resident Population by Sex, Age, Race, and Hispanic Origin for the United States and States: April 1, 2010 to July 1, 2017	2017 Population Estimates	?
PEPSRSH	Annual Estimates of the Resident Population by Sex, Age, Race, and Hispanic Origin for the United States, States, and Counties: April 1, 2010 to July 1, 2017	2017 Population Estimates	?
PEPSYSEX	Annual Estimates of the Resident Population by Single Year of Age and Sex for the United States, States, and Puerto Rico Commonwealth: April 1, 2010 to July 1, 2017	2017 Population Estimates	?
GCT-PEPANNCHG	Estimates of Resident Population Change and Rankings: July 1, 2016 to July 1, 2017 - State - County / County Equivalent	2017 Population Estimates	?
GCT-PEPCMCHG	Cumulative Estimates of Resident Population Change and Rankings: April 1, 2010 to July 1, 2017 - State - County / County Equivalent	2017 Population Estimates	?

Prepare Census Data

- ▶ Open the PEP_2017_PEPANNRES_with_ann CSV file
- ▶ Delete the top row (variable names), and keep only the columns “id2, Geography, and Population Estimate (as of July 1)–2017”
- ▶ Rename the final column “Population”
- ▶ Save the spreadsheet using the name “Population_2017”

A	B	C	D	E	F	G	H	I	J	K	L	M
1	GEO.id2	GEO.displayname	resbase420	resbase420	respop720	respop720	respop720	respop720	respop720	respop720	respop720	respop720
2	id2	Geography	April 1, 2017	April 1, 2017	Population	Population	Population	Population	Population	Population	Population	Population
3	0400000U	1 Alabama	4779736	4780135	4785579	4798649	4813946	4827660	4840037	4850858	4860545	4874747
4	0400000U	2 Alaska	710231	710249	714015	722259	730825	736760	736759	737979	741522	739795
5	0400000U	4 Arizona	6392017	6392309	6407002	6465488	6544211	6616124	6706435	6802262	6908642	7016270
6	0400000U	5 Arkansas	2915918	2916031	2921737	2938640	2949208	2956780	2964800	2975626	2988231	3004279
7	0400000U	6 California	37253956	37254518	37327690	37672654	38019006	38347383	38701278	39032444	39296476	39536653
8	0400000U	8 Colorado	5029196	5029325	5048029	5116411	5186330	5262556	5342311	5440445	5530105	5607154
9	0400000U	9 Connecticut	3574097	3574114	3580171	3591927	3597705	3602470	3600188	3593862	3587685	3588184
10	0400000U	10 Delaware	897934	897936	899712	907884	916868	925114	934805	944107	952698	961939
11	0400000U	11 District of Columbia	601723	601766	605040	620336	635630	650114	660797	672736	684336	693972
12	0400000U	12 Florida	18801310	18804594	18846481	19097369	19341327	19584927	19897747	20268567	20656589	20984400
13	0400000U	13 Georgia	9687653	9688690	9712696	9810595	9911171	9981773	10083850	10199533	10313620	10429379
14	0400000U	15 Hawaii	1360301	1360301	1363817	1378323	1392772	1408038	1417710	1426320	1428683	1427538
15	0400000U	16 Idaho	1567582	1567650	1570912	1583180	1594673	1610187	1630391	1649324	1680026	1716943
16	0400000U	17 Illinois	12830632	12831565	12841196	12862298	12878494	12890403	12882438	12862051	12835726	12802023
17	0400000U	18 Indiana	6483802	6484125	6490029	6515358	6535665	6567484	6593182	6610596	6634007	6666818
18	0400000U	19 Iowa	3046355	3046869	3050223	3063690	3074386	3089876	3105563	3118473	3130869	3145711
19	0400000U	20 Kansas	2853118	2853130	2858403	2868756	2885316	2892900	2899553	2905789	2907731	2913123
20	0400000U	21 Kentucky	4339367	4339340	4347948	4368505	4383673	4399121	4410415	4422057	4436113	4454189
21	0400000U	22 Louisiana	4533372	4533478	4544871	4574388	4602681	4626795	4648797	4671211	4686157	4684333
22	0400000U	23 Maine	1328361	1328362	1327568	1327968	1328101	1327975	1328903	1327787	1330232	1335907
23	0400000U	24 Maryland	5773552	5773807	5788099	5843115	5891680	5932654	5970245	6000561	6024752	6052177
24	0400000U	25 Massachusetts	6547629	6547808	6564943	6612178	6659627	6711138	6757925	6794002	6823721	6859819
25	0400000U	26 Michigan	9883640	9884129	9876731	9876199	9886610	9899219	9914675	9918170	9933445	9962311
26	0400000U	27 Minnesota	5303925	5303924	5310711	5345967	5377695	5416074	5452649	5483238	5525050	5576606
27	0400000U	28 Mississippi	2967297	2968103	2970437	2977452	2982963	2987721	2988578	2985297	2985415	2984100
28	0400000U	29 Missouri	5988927	5988925	5995681	6010280	6023267	6041142	6058014	6072640	6091176	6113532



A	B	C	D	E
1	id2	Geography	Population	
2	1	Alabama	4874747	
3	2	Alaska	739795	
4	4	Arizona	7016270	
5	5	Arkansas	3004279	
6	6	California	39536653	
7	8	Colorado	5607154	
8	9	Connecticut	3588184	
9	10	Delaware	961939	
10	11	District of Columbia	693972	
11	12	Florida	20984400	
12	13	Georgia	10429379	
13	15	Hawaii	1427538	
14	16	Idaho	1716943	
15	17	Illinois	12802023	
16	18	Indiana	6666818	
17	19	Iowa	3145711	
18	20	Kansas	2913123	
19	21	Kentucky	4454189	
20	22	Louisiana	4684333	
21	23	Maine	1335907	
22	24	Maryland	6052177	
23	25	Massachusetts	6859819	
24	26	Michigan	9962311	
25	27	Minnesota	5576606	
26	28	Mississippi	2984100	
27	29	Missouri	6113532	
28	30	Montana	1050493	

Prepared for QGIS Exercise

- ▶ Store the shapefile folder and the population folder under one larger folder
- ▶ For this exercise, the naming convention is less important
 - ▶ However, it is a good habit to name all folders and files without spaces, numbers, or punctuation
 - ▶ This becomes critical when doing more advanced spatial work



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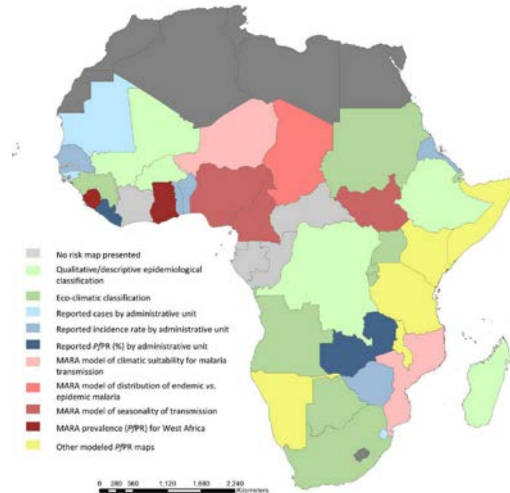
Exercise Using QGIS

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Choropleth Map

- ▶ For this exercise, we will make a choropleth map showing the population gradient across the US at the state level
- ▶ A choropleth map is a thematic map that depicts various quantities (or differences between quantities) by using shading or patterning
- ▶ This uses given geographic boundaries (i.e., state lines)

Choropleth map



Heat map

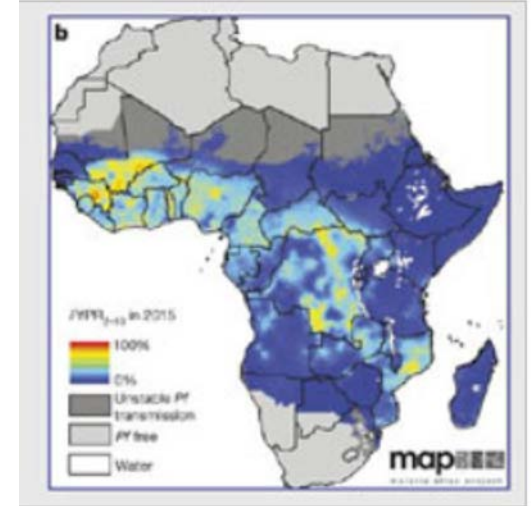
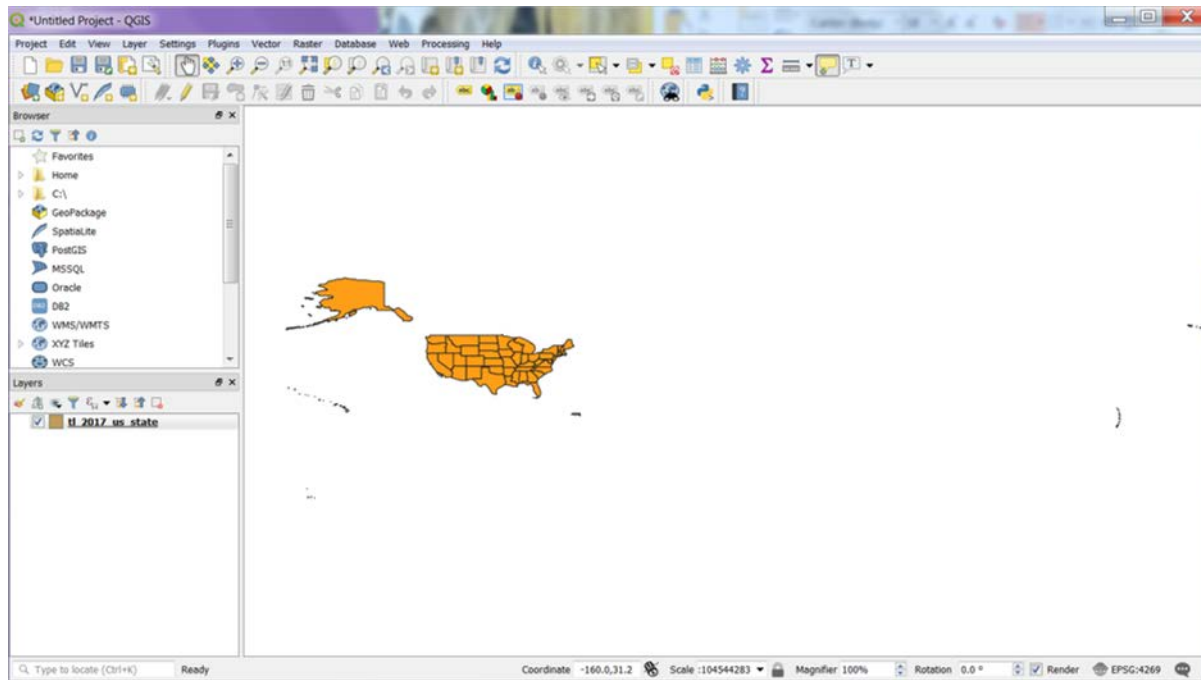


Image source: Choropleth map: Omumbo, J. A., Noor, A. M., Fall, I. S., & Snow, R. W. (2013). How well are malaria maps used to design and finance malaria control in Africa? *PLoS One*. doi: [10.1371/journal.pone.0053198](https://doi.org/10.1371/journal.pone.0053198). Heat map: Potash, S. (2016). Dr. Chris Murray discusses findings and future of Global Burden of Disease study at NIH. NIH Fogarty International Center. Retrieved October 10, 2018.

QGIS: Let's Get Started

- ▶ Open QGIS
- ▶ Under “Project,” select “New”
- ▶ Under “Layer,” select “Add Layer,” “Add Vector Layer,” and navigate to your US state shapefile
- ▶ Select the .shp file to add

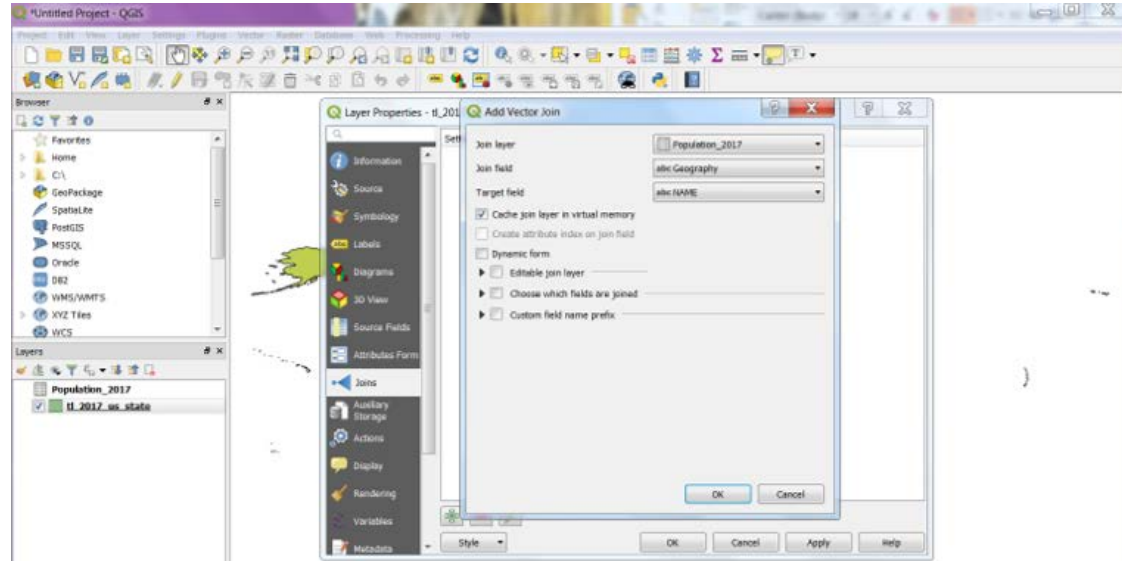


Add in Population Data

- ▶ Under “Layer,” select “Add Layer,” “Add delimited text layer”
- ▶ Navigate to your population CSV file
- ▶ Ensure that “First Record has field names” is selected under “Record and fields options”
- ▶ Select “No geometry (attribute only table)” under “Geometry definition”
- ▶ The CSV file should appear in the “Layers” window along with your shapefile

Join in Population Data

- ▶ Right-click the shapefile and select “Properties”
- ▶ Go to “Joins” and select the green plus sign button
- ▶ In the dialog box, select your population file as the “Join layer,” “Geography” as the “Join field,” and “Name” as the “Target field”
- ▶ Click “OK” and apply the join

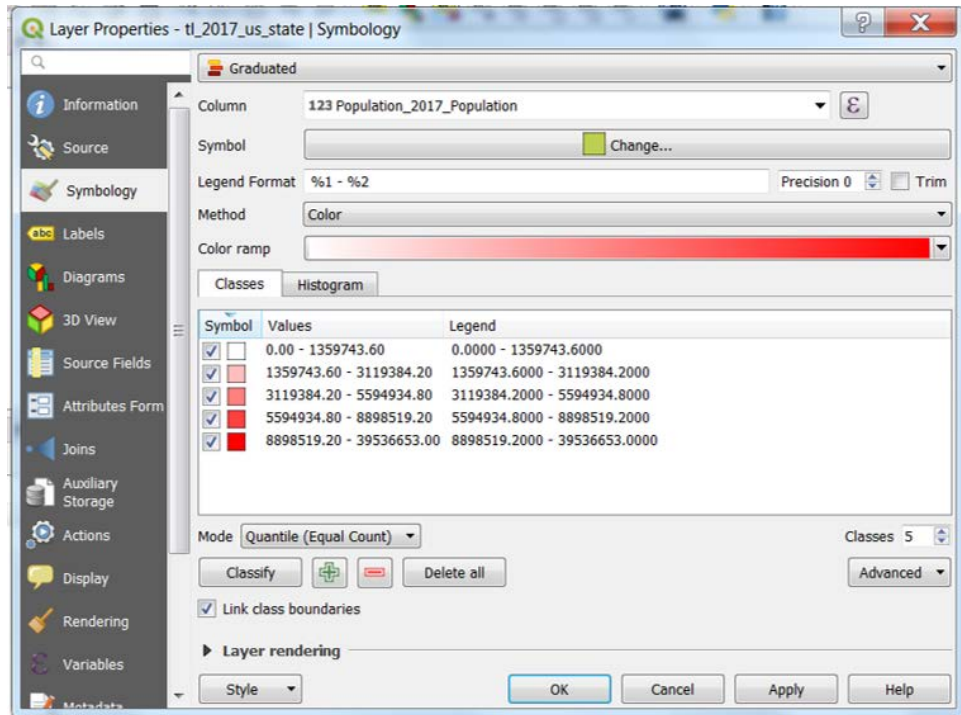


Create Graduated Categories

- ▶ Open your attribute table by right-clicking on the shapefile and selecting “Open Attribute Table”
 - ▶ You should see the population values as the final column
- ▶ Right-click on the shapefile and select “Properties,” then “Symbology”
- ▶ Select “Graduated” in the top pull-down menu
 - ▶ Under “Column,” select “Population”
- ▶ Under “Mode,” select “Quantiles”
 - ▶ This tells the program to put an equal number of states in each of however many “Classes” you choose
 - ▶ This is considered the optimal method for displaying epidemiologic data in choropleth maps

Modify the Layer

- ▶ You may now select the number classes, color scheme, and legend format of your choosing
- ▶ Apply your changes and click “OK”



Label the States

- ▶ Right-click the shapefile layer and go to “Properties”
- ▶ Select “Labels”
- ▶ Select “Single labels” in the top drop-down menu, and then select “Name” as the column for “Label with”
- ▶ Apply the labels
- ▶ You can modify the font and font size as you please

Save the Map Project

- ▶ Ideally, a map should be easy to read and informative as a stand-alone document
- ▶ The next steps will be to create such a map
- ▶ Click “Zoom in” under “View” and zoom into the area depicting the 50 United States and Puerto Rico
- ▶ Save this project as “Population_Map”

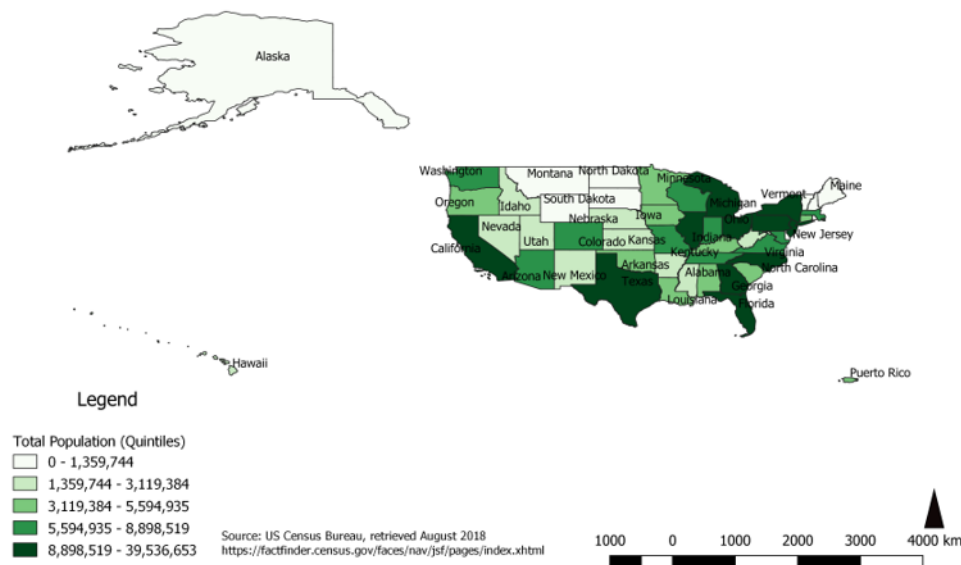
Create and Exportable Map

- ▶ Under “Project,” select “New Print Layout”
- ▶ Name this file “Population_Map”
- ▶ Under “Add Item,” select “Add Map” and then draw a box in the layout area in which your map will appear
 - ▶ You can resize and move this map around as you choose
- ▶ Add components to make your map interpretable, such as a legend, scale bar, title, etc.
 - ▶ These things can be done under “Add Item”
- ▶ Once you are finished, go under “Layout” to export your map into whatever format you choose

Learning More about QGIS

- ▶ You will have a product that looks something like this map
- ▶ There are numerous QGIS online tutorials for those interested in further refining mapmaking skills
- ▶ In addition, there are many tutorials to learn more about how a map should ideally look

US Total Population by State, 2017



Summary

- ▶ Our environment has a significant influence on our health and well-being
- ▶ GIS is a tool for integrating and visualizing spatial data from multiple sources
- ▶ The US Census provides geographic shapefiles as well as demographic data that is very useful for population health assessment and monitoring
- ▶ A choropleth map is a thematic map that depicts various quantities (or differences between quantities) by using shading or patterning

Thank you!